Plan-Validity-Driven Replanning Under a Fixed Planning Budget

Abstract

A hierarchical controller must decide not only what plan to execute but also when to replace it. We study whether a prediction of the value of renewing the current plan can improve that timing without increasing the number of planner calls. A goal-conditioned controller is first trained with variable option durations. Paired simulator branches then supply supervision for keeping or renewing its current waypoint. A predictor using causal plan-state and motion history features triggers the first threshold crossing within each planning bin, with a deadline that preserves exactly one call per bin. In a frozen PointMaze hidden-regime confirmation with eight independent optimizer seeds, the trigger reduced episode tracking-error integral by 0.506 (95% CI 0.408-0.603) and increased return by 40.66 (31.55-49.78) relative to periodic planning, with 24 planner calls in every episode. A separate 512-path PointMaze evaluation on eight existing trained roots found 5.81% lower mean tracking error integral than a selected distance threshold and 21.56% lower than a selection-based timing portfolio under shared compute caps. Two applicability screens separate the usefulness of fresh waypoints from the usefulness of changing their timing: Reacher benefited from fresh waypoints without a reliable gain from the tested event schedule, while a cart-pole tracking task showed timing gains but controller-dependent benefits of waypoint freshness. Separate Transit studies reproduced single-intervention gains, but a 320-scene two-decision evaluation did not confirm joint cost and waiting improvement. The positive timing result is confined to the PointMaze task family.

1. Introduction

Planning and execution operate on different clocks. A lower controller can continue to respond to physical feedback while a higher-level waypoint remains unchanged. Periodic planning makes this division simple, but assigns the same refresh schedule to a useful plan and an obsolete one. The decision studied here is whether the present plan is worth replacing, conditional on the controller that will execute the replacement.

The motivating Freq-HRL scope is multiscale time-series control with persistent high-level plans and fast closed-loop execution. It is an architectural protocol that can use PPO or SAC as a learning backend, rather than an alternative general-purpose RL optimizer. The present study tests its renewal-timing question within this planning/execution class, not universal algorithm superiority.

Temporal abstraction already supports learned termination, action repetition and adaptive high-level decision frequency (Sutton et al. 1999; Bacon et al. 2017; Sharma et al. 2017; Zhou and Yu 2020). Our question is narrower: can direct supervision of a keep-versus-renew value difference improve the allocation of a fixed number of upper calls? Separating this timing decision from controller learning makes it possible to compare schedules using exactly the same trained upper and lower policies.

The proposed procedure learns an operational plan-validity predictor. Its target is a paired short-horizon tracking loss difference, obtained by replaying the same simulator prefix while either retaining the waypoint or requesting a new one. At deployment, the predictor uses only current and past observations. It can move a call within a predefined bin but cannot add calls. Full-episode evaluation tests whether the local supervision remains useful after the resulting decisions change the states subsequently visited.

The study makes three contributions. First, it specifies a reproducible branch-supervised timing rule with an exact planner-call constraint. Second, it reports a frozen independent-seed PointMaze confirmation against

periodic, random-offset and current-only timing controls, and a separate comparison with stronger timing controls on fresh paths. Third, it tests two conditions for applicability: whether updating the plan helps at a matched clock, and whether changing call timing helps at a matched budget. These tests distinguish two failure modes. A separate Transit extension tests whether benefits measured for one intervention also appear under repeated control.

2. Related Work

Temporal abstraction. Options formalize extended actions, and Option-Critic learns intra-option policies and termination conditions (Sutton et al. 1999; Bacon et al. 2017). Our controller is trained with semi-Markov upper transitions; the added timing predictor is fitted afterwards from paired branches.

Adaptive decision timing. FiGAR learns action repetition lengths, TEMPLE adapts high-level decision frequency, and TAAC chooses between a previous and a newly proposed physical action (Sharma et al. 2017; Zhou and Yu 2020; Yu et al. 2021). These are direct conceptual neighbors. Here the executed lower action remains closed-loop at every primitive step, while the upper waypoint refresh is restricted to one call per bin. The present comparisons isolate timing effects under a shared trained controller.

Allocating planning computation. Learning when to plan also appears in language-model agents, including a supervised-plus-RL approach to dynamic planning in Crafter (Paglieri et al. 2025). Our setting uses continuous-control waypoints, paired simulator supervision and a fixed call-count contract. The present comparisons establish schedule utility, not a general advantage in test-time compute allocation.

3. Method

3.1 Controller and timing problem

Let h_t denote the observable history at primitive time t, and let g_t be the current waypoint. An upper policy pi_U produces a relative state-space goal; a goal-conditioned lower policy pi_L produces every physical actuator action. Both policies are learned with PPO (Schulman et al. 2017). An upper transition spans the actual interval between waypoint changes, and lower intrinsic GAE ends at the realized waypoint boundary.

The episode has T = 1200 steps of 0.01 seconds. With K = 50, bin b contains steps bK through (b+1)K-1. The timing rule must make exactly one upper call per bin. The first call occurs at step zero. Subsequent bins allow calls only at offsets {0, 5, 10, 15, 20, 25}; the waypoint is held until the next actual call. The variable inter-call durations are therefore 25-75 steps. This is a constrained within-bin timing problem, not an unconstrained termination process.

3.2 Variable-duration controller training

Each optimizer seed uses eight controller-training paths and eight separate checkpoint-selection paths. Training lasts 384 iterations. A seeded random-offset schedule makes one upper call per bin, with offsets from zero to 25, so the controller encounters variable durations before trigger fitting. The selected upper/lower weights are then frozen across all four timing modes. The controller has 267,018 parameters; it uses a 0.64-second observation history, a 0.50-second nominal upper period and a 0.04-second fast feature timescale.

3.3 Paired renewal supervision

On eight separate branch-fit paths, a query is replayed twice from the same prefix: keep retains the current waypoint; renew invokes the upper policy once at the query. Both branches run the lower controller for the same next H = 50 steps without further upper calls. The target is

$$\Delta_H(h_t, g_t) = L_H^{\text{keep}}(h_t, g_t) - L_H^{\text{renew}}(h_t, g_t), \qquad L_H = 0.01 \sum_{j=0}^{H-1} \|e_{t+j}\|_2^2,$$

where e is the task tracking error. A positive value favors renewing for this branch horizon. Prefix and feature equality checks pair the interventions; the lower controller continues to respond to the evolving physical state.

Each path supplies four queries from each of six simulator-defined classes: three lags after a regime change, a force pulse, a distractor change and a matched neutral opportunity. This gives 192 paired queries per optimizer seed. Event information stratifies offline query acquisition, including neutral-window selection. It is excluded from predictor inputs and deployment decisions.

3.4 Causal-history predictor

The candidate uses 39 features of plan age, tracking and waypoint geometry, observed target velocities over 1/10/25/50-step lags, force magnitude, and their fixed interactions. The current-only control uses 170 quadratic features constructed from 17 current state/plan variables. Both models standardize features using fit data and solve ridge regression with an unpenalized intercept. The ridge coefficient is selected by leave-one-fit-path-out MSE from {0.01, 0.1, 1, 10, 100, 1000, 10000}.

For the selected coefficient, leave-one-fit-path-out predictions determine the 75th-percentile threshold q. This threshold is an empirical timing parameter, not a calibrated probability or a certificate that renewal has positive value. Evaluation paths do not select coefficients, features or thresholds.

3.5 Budgeted deployment

At an unused call opportunity in a bin, the predictor scores the current observable history and plan. The upper policy is called at the first score at least q, or at offset 25 if no earlier opportunity passes. Computing a predictor score does not call the upper policy. The lower policy acts at every step, including while the timing rule is waiting.

```
Fit and select the variable-duration controller on disjoint path sets.
Fit the renewal predictor and its threshold on branch-fit paths only.
Freeze controller weights, predictor and threshold.
For each held-out episode:
    Call the upper policy at step zero.
    In each later 50-step bin:
        At offsets 0, 5, 10, 15, 20, 25, until one call occurs:
            Score the currently observed plan and history.
            Renew if score >= threshold or offset == 25.
    Execute the lower policy on every primitive step.
```

Budget property. Every bin makes at most one call because later checks are disabled after renewal. Every bin makes at least one because offset 25 is a mandatory deadline. Thus every complete episode makes exactly T/K = 24 calls. This scheduling property does not imply optimal timing or lower total computational cost.

4. Experimental Design

4.1 Independent PointMaze confirmation

The hidden-regime task combines moving targets, physical force pulses and distractor observations. The frozen algorithm was developed before confirming it on optimizer roots 209011, 209023, 209037, 209049, 209061,

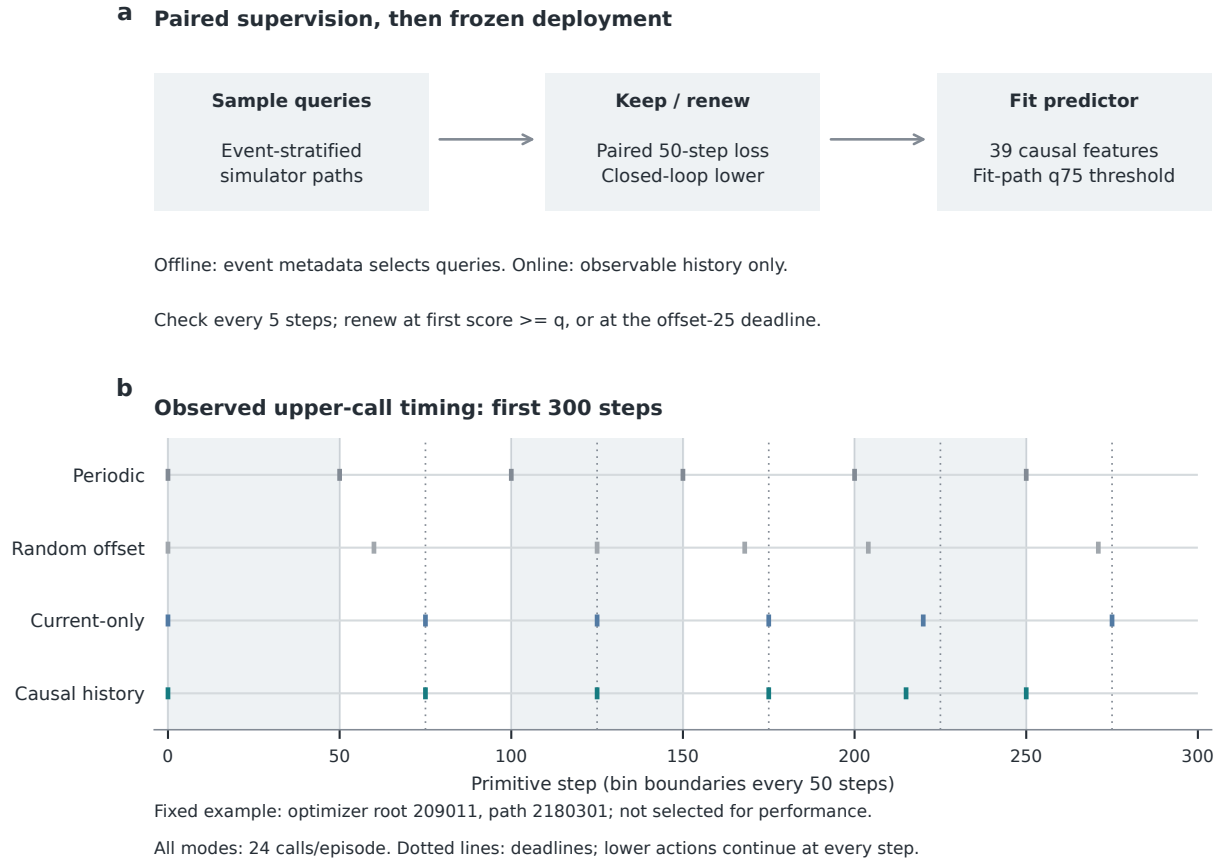


Figure 1: **Supervision and budgeted deployment.** (a) Event-stratified offline queries provide paired 50-step keep/renew losses for the 39-feature predictor. Deployment checks every five steps within the allowed offset window, with one upper call at the first threshold crossing or the offset-25 deadline. (b) The first 300 steps of the first registered confirmation root/path (209011/2180301) show actual saved upper-call times for all four modes; this example was not chosen for performance. Shading alternates 50-step bins; dotted lines mark deadlines after the initial bin. Lower actions remain closed-loop every step. All complete episodes have 24 upper calls. Source data.

209073, 209089 and 209101. Each root uses 8 training, 8 selection, 8 branch-fit and 16 trigger-evaluation path seeds. All 320 path seeds across these roles are distinct and disjoint from development.

Four timing modes share a root's trained controller and its evaluation paths: periodic calls at bin starts, seeded random-offset calls, the current-only trigger, and the causal-history candidate. The primary endpoint is fixed minus candidate episode integrated squared tracking error (ISE). Four additional registered gates test ISE against random-offset and current-only controls, return against fixed timing, and controller learning against initialization.

For each contrast, path differences are averaged within a root; the eight root means supply the two-sided 95% Student-t interval with seven degrees of freedom. The joint registered decision requires all five lower bounds to be positive. The 128 evaluation paths and 512 paired-mode rollouts do not replace the eight optimizer seeds as the independent units. Development and confirmation estimates remain separate.

4.2 Native Transit boundary study

Transit evaluates a different, finite-Monte-Carlo action-selection procedure. Independent P8 proposal and S8 screening futures support a service-aware timetable intervention with a frozen lower controller. It searches alternative timetable actions, rather than only timing a call to the PointMaze policy. It therefore tests an extension of the decision-value idea, not the same algorithm in a second domain.

Single-intervention studies use independent evaluation futures after actions and matching weights are sealed. Prefix-bootstrap intervals quantify scenario variation; paired conditional-MC intervals quantify future sampling at those prefixes. The subsequent two-decision study evaluates 320 fresh complete scenes, with four paired controllers, at upper callbacks 8 and 9. Each controller rebuilds its second state and P8/S8 menus after its own executed first action. Both its stratified bootstrap and paired normal intervals use complete scene/episode units; neither is a conditional-future MC interval.

4.3 Applicability screens

We test two operational questions separately. At the same fixed clock, does executing the current waypoint rather than the previous call's waypoint reduce ISE? With the same 24 calls, does an event schedule reduce ISE relative to a fixed clock selected on separate paths and a rotation of the event schedule's offsets? Rotation preserves the number of calls and the offset histogram while breaking alignment with events. The first contrast tests waypoint freshness under a fixed lower controller; the second tests a specified source of timing opportunity, not all possible schedules.

Reacher reuses two frozen moving-target pilot controllers and aligns calls to hidden regime events as a privileged probe. The derived cart-pole task tracks a visible position command using native plant dynamics and fixed LQR; it calls on command changes. Each screen uses 24 paired development paths per root or configuration. Neither fits or evaluates a new Stage9 predictor. The cart-pole follow-up crosses both prespecified LQR position weights with both fixed offsets on the same 32 fresh paths to separate the two changes.

5. Results

5.1 Plan-validity timing improved held-out PointMaze control

All five registered confirmation gates passed (Table 1). Every optimizer root had positive candidate-versus-fixed ISE and return effects. Mean episode ISE was 1.447072 with fixed timing, 1.549716 with random offsets, 1.202303 with the current-only trigger and 0.941530 with the causal-history trigger. The candidate's 34.94% reduction relative to fixed timing is a ratio of pooled point means, not an interval estimate for a percentage effect.

Table 1: **Frozen independent-seed PointMaze confirmation.** Positive contrasts favor the candidate or trained controller. Intervals use eight paired root means.

Registered contrast	Mean	95% CI
Fixed - candidate episode ISE	0.505541	[0.408443, 0.602640]
Random-offset - candidate episode ISE	0.608186	[0.570111, 0.646260]
Current-only - candidate episode ISE	0.260773	[0.152216, 0.369331]
Candidate - fixed episode return	40.662456	[31.548406, 49.776506]
Untrained - trained controller ISE	20.862987	[15.939428, 25.786547]

All modes made exactly 24 upper calls per episode. The candidate made 13.1953 pre-deadline calls on average, showing that it did not simply wait for every deadline. Its mean 84.1875 score checks per episode (range 48-120) were additional inference operations. The eight-root branch-fit phase replayed 1,942,606 primitive steps, including repeated prefixes; controller training and selection are separate costs. These counts establish call parity, not equal FLOPs or a measured runtime advantage.

The earlier eight-root development result also passed, with ISE advantage 0.396372 [0.275784, 0.516959] and return advantage 30.770657 [21.111415, 40.429899] against fixed timing. These are supporting development observations, not another eight roots added to the confirmation interval.

Comparison with a selected distance threshold. In a separately registered development diagnostic after the strong-comparator build gate stopped formal evaluation, all eight completed controller builds were retained. The Stage9 rule and the per-root selected distance threshold ran on the same 32 new paths per root, with 24 upper calls each. Mean episode ISE was 1.01091 versus 1.09029 (7.28% lower). The paired optimizer-root advantage was 0.07938 (nominal 95% t interval [0.03008, 0.12868]); all eight root means and both path halves favored Stage9. One learned termination rule had failed the best-clock competence gate before test access, so this result addresses the distance threshold only and is separate from the original confirmation. Diagnostic source.

Comparison with a timing portfolio. Before accessing 64 further paths per root, the comparator was frozen to whichever of learned termination and best constant clock had lower ISE on that root's old timing-selection paths (ties to the clock). Seven roots selected termination and one selected the clock. Both searches were charged. The same eight controllers and all paths were paired across methods; every episode used 24 calls.

Table 2: **PointMaze timing portfolio and distance threshold.** Positive gains favor the Stage9 timing rule.

Comparator	Episode ISE (Stage9: 0.99934)	Paired root ISE gain [nominal 95% CI]	Relative gain
Selected distance threshold	1.06097	0.06163 [0.01980, 0.10345]	5.81%
Selected termination or clock	1.27402	0.27467 [0.19309, 0.35625]	21.56%

Both path halves and the joint registered gate passed. Stage9 beat the portfolio on all eight roots and the

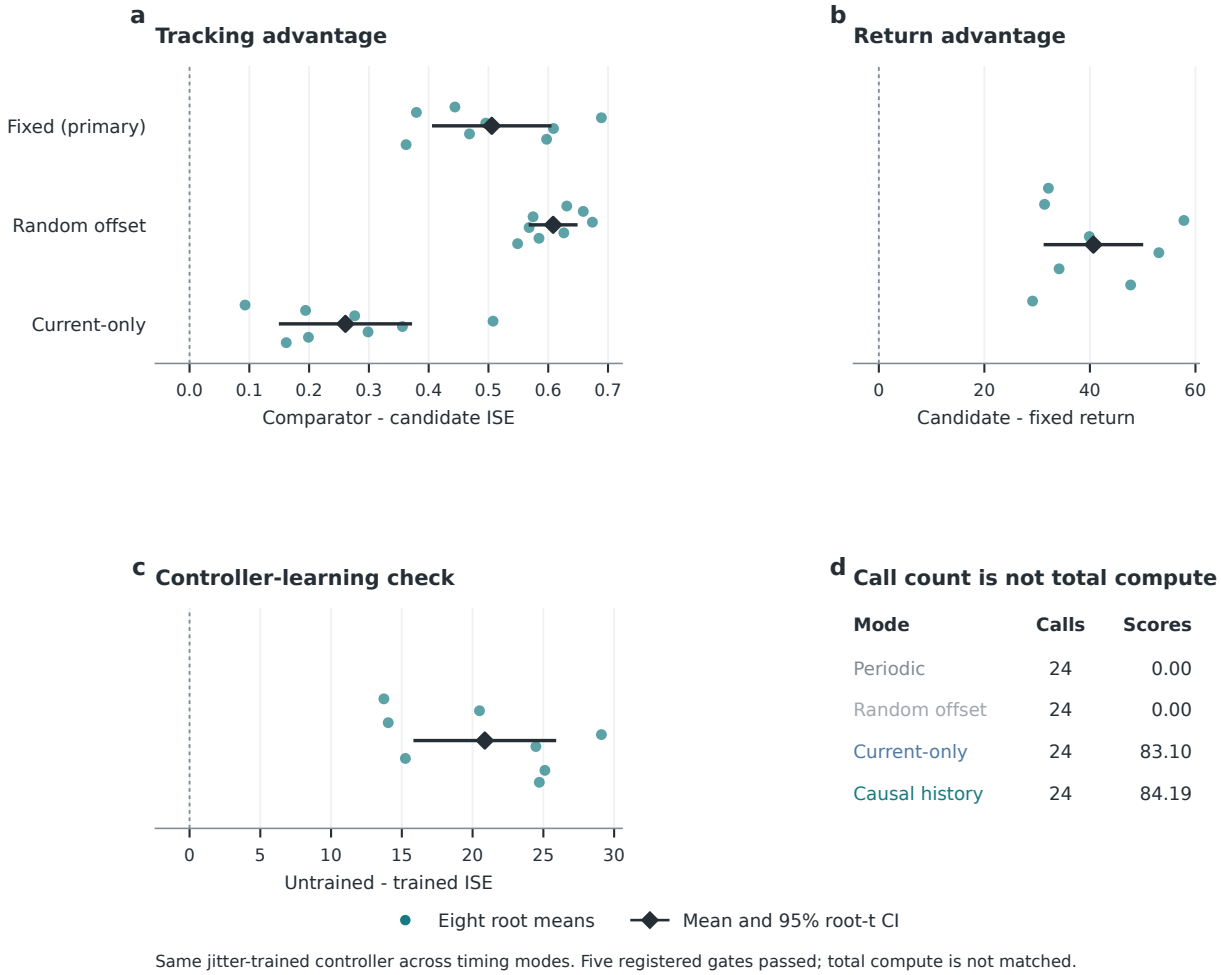


Figure 2: **Independent-seed PointMaze confirmation.** (a) Episode ISE advantages over fixed (primary), random-offset and current-only timing; (b) return advantage over fixed timing; (c) the required learning check, comparing untrained with trained controller ISE under canonical random-offset deployment. Teal dots are eight optimizer-root means, averaging 16 evaluation paths each; black diamonds and lines are equal-root means and two-sided 95% Student-t intervals (df7), reproducing Table 1's five registered gates. All lower bounds are positive. These are the original per-endpoint intervals, not simultaneous or research-wide corrected intervals. (d) All modes use 24 upper calls; current-only and history triggers additionally make 83.10 and 84.19 score checks per episode on average. Shared jitter-trained weights isolate schedule effects; call parity does not establish equal total compute. Source data.

distance threshold on seven. All methods met the 10,000 CPU-second lifecycle, 6.5-million offline-step and 10ms control p99 caps. These are fresh paths, not another independent training cohort. Portfolio result and accounting.

Matched timing features. Three heads shared 12 inputs, 91 quadratic coefficients including the intercept, slope effective DF 8, and identical branch labels and OOF calibration. On 512 further paired paths, the 100ms target-velocity/force-history head reduced ISE by 6.44% versus the current-only view (root gain 0.07906, nominal 95% CI [0.01870, 0.13942]). Changing only the target-velocity horizon to 500ms worsened ISE by 5.85% versus the 100ms head, with both path halves worse. The original Stage9 rule remains the best observed anchor on these paths, but differs in features and fitting. The ablation supports short-history utility, not a multiscale attribution of the original advantage. Ablation report.

5.2 Historical learned-upper gains are real but method-specific

Later PointMaze studies established additional learned-upper effects under their own contracts. Fresh-sample Stage88 learning improved joint-call return over lower-only by 0.2215/0.4801 at periods50/100; actor swaps isolated an upper contribution. Stage100 replicated staged upper gains on eight new teacher roots, although most route-level advantage was inherited lower quality. A compact fixed-lower upper update in Stage135 improved over causal forecast by 1.985835 [1.490466, 2.363561] and 1.079190 [0.679043, 1.333611], using within-stage multiplicity-corrected root intervals. These gains were only 0.1724% and 0.0947% of forecast return. The following ordinary joint PPO update lost the warm-start advantage in all four diagnostic root/period cells.

These studies reject the blanket interpretation that a learned upper never changed useful behavior. They do not constitute independent replications of the timing predictor in Table 1. A full historical evidence inventory is retained in the supplementary source map.

5.3 Single-intervention Transit benefits replicated on new prefixes

Two separate 400-prefix cohorts reproduced negative total-cost and waiting contrasts for the frozen joint selector against raw control. The second also evaluated a service-only selector, which improved both endpoints under both interval types and in both evaluation halves. Table 3 reports their scope without pooling cohorts or treating fixed historical profiles as training seeds.

Table 3: **Native Transit single-intervention outcomes.** Negative favors the first controller; cost and waiting/minute values are multiplied by 1e6.

Study / contrast / endpoint	Mean	Prefix CI95	Conditional MC CI95
Stage298 / joint - raw / cost	-202.006	[-288.844, -119.627]	[-265.551, -138.460]
Stage298 / joint - raw / wait	-1182.737	[-1658.035, -781.291]	[-1299.168, -1066.305]
Stage301 / service-only - raw / cost	-318.120	[-424.992, -215.424]	[-396.701, -239.539]
Stage301 / service-only - raw / wait	-1230.467	[-1757.211, -825.834]	[-1357.549, -1103.385]
Stage301 / joint - raw / cost	-307.139	[-403.590, -216.799]	[-376.580, -237.698]
Stage301 / joint - raw / wait	-1070.902	[-1583.164, -673.425]	[-1185.750, -956.054]
Stage301 / joint - service-only / cost	10.981	[-38.466, 60.105]	[-26.841, 48.802]
Stage301 / joint - service-only / wait	159.565	[82.814, 250.561]	[110.377, 208.754]

The Stage301 service-only average waiting reduction was approximately 0.074 seconds per single intervention. Joint selection had no confirmed extra cost benefit over service-only and significantly worsened waiting. Better-than-raw performance therefore did not establish the value of the additional fleet critic.

5.4 The two-decision Transit confirmation failed its joint criterion

All 1,280 controller evaluations completed successfully across 320 paired scenes. Including inner sampling and identity replays, 219,520 native episodes passed the execution ledger. The service-only-versus-raw primary did not pass: total cost increased at the point estimate, waiting had a crossing normal interval, and the two half-cohort cost effects had opposite signs (Table 4).

Table 4: **Complete-episode Transit boundary.** Negative favors the first controller. Values use the same 1e6 scaling as Table 3. Both intervals here measure complete-episode sampling, not the two uncertainty sources in Table 3.

Contrast / endpoint	Mean	Episode bootstrap CI95	Paired normal CI95
service-only - raw / cost	1901.331	[-2729.400, 6937.962]	[-2856.632, 6659.294]
service-only - raw / service component	-702.835	[-1497.065, -120.823]	[-1414.375, 8.704]
service-only - raw / wait	-5706.250	[-12519.141, -787.422]	[-11818.256, 405.756]
joint - service-only / cost	28.572	[-5.247, 84.056]	[-19.407, 76.551]

Service-only cost was +4109.200 in the first 16-scene half per stratum and -306.537 in the last; waiting was -9462.500 and -1950.000. The pooled and half-cohort point waiting-retention ratios were 89.51%, 93.57% and 73.93% of full base. However, the paired margin comparing each effect with half of base did not have negative upper bounds under both interval methods in all three parts. The statistical 50% retention criterion therefore failed. Joint likewise failed transfer and added-cost-value criteria.

The fleet component explains the pooled cost sign: its mean penalty +0.002604167 exceeded the service saving -0.000702835. Four scenes changed peak fleet, with three increases and one decrease, all in the first half. Joint and service-only produced the same outcomes in 315/320 scenes and the same peak fleet in all 320. The risk filter did not change those fleet events. Service-only changed outcomes in only 21/320 scenes; its largest waiting-saving scene supplied 46.66% of net wait benefit while increasing total cost. These event and concentration counts are descriptive, not additional significance tests.

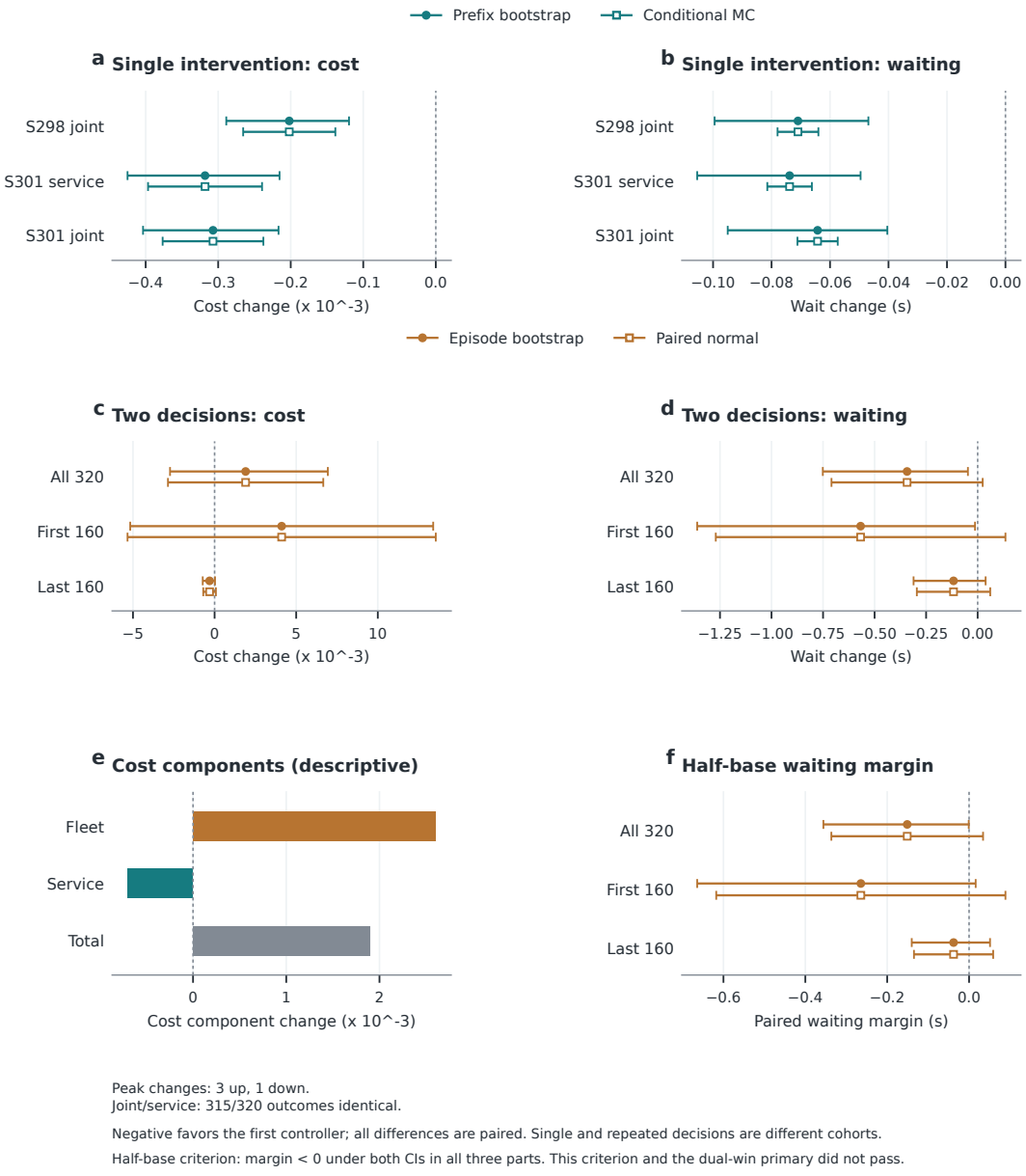


Figure 3: **Local Transit gains and the repeated-decision boundary.** (a,b) Frozen single-intervention contrasts against raw control on separate 400-prefix Stage298/301 cohorts: circles show stratified prefix-bootstrap CI95 and squares paired conditional-MC CI95. Fixed profiles and queried positions are averaged within prefixes, not counted as independent training seeds. (c,d) Stage304 service-only minus raw, pooled over 320 complete scenes and split into first/last 160-scene halves (16 scenes per root/regime stratum in each half): circles show episode-bootstrap CI95; squares show paired-normal CI95. Both intervals now measure complete-episode sampling. (e) Descriptive mean fleet, service and total-cost components; inference for total cost is in c. Three increased peaks and one decreased peak dominate the fleet difference. Joint and service-only outcomes coincide in 315/320 scenes, and the additional joint cost benefit is unconfirmed (Table 4). (f) Paired wait margin `delta_wait_service - 0.5 delta_wait_base`; retaining at least half of base's benefit requires negative upper bounds under both CIs in the pooled cohort and both halves. This criterion did not pass. Negative favors the first controller. Cost axes use units of 1e-3; waiting and margins are converted from minutes to seconds. These studies use different cohorts and an action-selection method different from the PointMaze trigger; no cross-panel pooling is performed. Source data.

5.5 Useful waypoints and useful timing are distinct

Table 5: **Applicability screens.** Positive entries are relative reductions in whole-episode ISE versus the comparator. Path-paired contrasts are computed within each row; rows are not pooled.

Frozen root or configuration	Current vs previous waypoint	Event vs selected clock	Event vs rotated offsets
Reacher 1972210000	+12.30%	-0.22%	+0.30%
Reacher 1972220000	+6.02%	+1.99%	+1.41%
Cart-pole, LQR position weight 10	-8.30%	+5.13%	+5.03%
Cart-pole, LQR position weight 30	+4.41%	+10.03%	+9.77%

In Reacher, waypoint gains persisted in the latter halves of options, but all four event timing intervals crossed zero. In the cart-pole screen, the visible trigger passed both timing comparisons in both configurations, but the waypoint effect reversed, failing the joint qualification.

The cart-pole follow-up reproduced the reversal at both fixed offsets. Relative ISE gains from current waypoints were -2.68% and -2.63% at weight 10, versus +7.08% and +7.41% at weight 30. The absolute freshness contrast increased by 0.02458 (nominal paired 95% CI [0.02299, 0.02618]) when changing the LQR weight; both the offset and interaction intervals crossed zero. Weight 30 nonetheless had higher absolute ISE at both offsets. The follow-up identifies a dependence on the execution policy; it does not identify a better controller or reverse the qualification decision.

6. Discussion

The main positive result concerns decision timing with a fixed learned controller. A plan-dependent value signal improved the use of an unchanged number of upper calls on fresh PointMaze optimizer seeds. The current-only control shows that this result was not explained by its particular current state/plan representation alone; it does not isolate history information from all feature-design or regularization effects.

The applicability screens explain why slow plans and fast execution are not enough on their own. A plan update can be useful without the tested timing signal improving its allocation. Conversely, useful timing can coexist with a lower controller that tracks better when given a lagged waypoint. The planner, lower controller and timing signal therefore need to be evaluated together. These screens are development evidence for that distinction.

The Transit extension distinguishes information value from a reliable risk decision. Its service-aware single-intervention gains survived independent futures and new prefixes, while adding a learned fleet score did not establish incremental cost value. In the two-decision study, the fleet filter left the few consequential peak events unchanged. More accurate average prediction or a passing software gate would not resolve this control-level result.

The broader research history also separates three objects: reducing a frequency-responsibility diagnostic, learning a useful upper action, and learning when to renew that action. The positive result for one object cannot be substituted for evidence about another. This distinction gives the present paper a coherent timing contribution without discarding the earlier findings.

7. Limitations

Task and baseline coverage. The confirmed timing mechanism has one hidden-regime PointMaze task family and eight fresh optimizer seeds. The 170-feature current-only comparator is not a capacity-matched history ablation. The matched-head study identifies a useful short-history view, but does not isolate the original Stage9 gain or establish a general benefit from mixing horizons. Later learned termination was active but one new root failed best-clock competence, so the learned-only evaluation was not run. The portfolio result compares against a selector, not every learned termination rule. Sharing jitter-trained controller weights isolates the effect of timing, but does not compare timing methods with controllers trained under each method's own schedule.

Supervision and cost. Training uses simulator branch access and event-stratified query acquisition, including privileged information for query sampling. Deployment does not use future observations or event labels. The procedure remains unvalidated with label-free or unstratified data acquisition. Planner-call parity excludes score checks, feature updates, branch simulation and policy-training cost. The portfolio study adds common caps with measured costs, not equal realized cost. Performance across different compute budgets remains untested.

Transfer and horizon. Neither applicability screen confirms the same learned timing rule on a second task. The Reacher probe does not rule out other timing signals, and the cart-pole result covers two fixed feedback configurations on a derived tracking task, not the stock benchmark. Transit uses a different action-search mechanism, two historical lower roots and, in the two-decision study, one fixed profile. Only two of 44 upper callbacks use the tested selector. Its comparison with the single-intervention cohorts changes profile composition, decision positions, on-policy states and scenes, so failure cannot be attributed uniquely to repeated decisions. It is not same-algorithm cross-domain confirmation or real-world passenger-demand deployment evidence.

Statistical and theoretical scope. Each study retains its frozen analysis; root, prefix, conditional-future and whole-episode intervals are distinct. The studies are not pooled into a research-wide significance claim. A local branch advantage neither bounds the value of first-threshold-crossing timing over an episode nor certifies fleet-cost safety. Exact call accounting is the only deterministic guarantee of the deployed timing rule established here.

8. Reproducibility

The timing algorithm was frozen at revision `91b3e6919bcffe3a75d4942b0df31ae8478d69f6`. Its executable implementation is `freq_hrl/experiments/pointmaze_budgeted_trigger.py`, with feature/branch definitions in `pointmaze_compact_plan_validity.py` and `pointmaze_plan_validity_branching.py`, and the registered constants in `scripts/pointmaze_budgeted_trigger_stage9_spec.py`.

The reproduction instructions and source map identify the executable code, registrations, compact per-episode PointMaze outcomes and saved Transit reports. From these inputs, the artifact recomputes PointMaze root effects and intervals, reproduces the reported Transit contrasts from the qualified summaries, and rebuilds the three figures and the manuscript PDF. Figure data are also available as CSV. This workflow uses no simulator, training, network connection or scheduler. The figure audit records the data and statistical units behind each figure.

Data and code availability. The accompanying artifact includes the inputs for the reported analyses, the manuscript and figure generators, and the frozen run specifications. The full repository contains the experiment implementations. Raw trajectories, server caches and checkpoints are not part of the artifact, so it does not independently rerun policy training or native simulations.

9. Conclusion

Plan-validity-driven timing improved PointMaze closed-loop tracking and return without increasing planner calls in a frozen independent-seed study. It also improved tracking against a distance threshold and a timing portfolio on fresh paths under shared compute caps. Reacher and cart-pole show why waypoint freshness and timing opportunity should be measured separately before transferring the method. The Transit result shows why local improvements need separate episode-level confirmation. The supported contribution is a timing rule for the evaluated planning and execution hierarchy, not a replacement for its learning backend.

References

Bacon, Pierre-Luc, Jean Harb, and Doina Precup. 2017. "The Option-Critic Architecture." *Proceedings of the Thirty-First AAAI Conference on Artificial Intelligence*. https://doi.org/10.1609/aaai.v31i1.10916.

Paglieri, Davide, Bartłomiej Cupiał, Jonathan Cook, et al. 2025. *Learning When to Plan: Efficiently Allocating Test-Time Compute for LLM Agents*. https://doi.org/10.48550/arXiv.2509.03581.

Schulman, John, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. 2017. *Proximal Policy Optimization Algorithms*. https://doi.org/10.48550/arXiv.1707.06347.

Sharma, Sahil, Aravind S. Lakshminarayanan, and Balaraman Ravindran. 2017. "Learning to Repeat: Fine Grained Action Repetition for Deep Reinforcement Learning." *International Conference on Learning Representations*. https://openreview.net/forum?id=B1GOWV5eg.

Sutton, Richard S., Doina Precup, and Satinder Singh. 1999. "Between MDPs and Semi-MDPs: A Framework for Temporal Abstraction in Reinforcement Learning." *Artificial Intelligence* 112 (1–2): 181–211. https://doi.org/10.1016/S0004-3702(99)00052-1.

Yu, Haonan, Wei Xu, and Haichao Zhang. 2021. "TAAC: Temporally Abstract Actor-Critic for Continuous Control." *Advances in Neural Information Processing Systems* 34: 29021–33. https://proceedings.neurips.cc/paper/2021/hash/f337d999d9ad116a7b4f3d409fcc6480-Abstract.html.

Zhou, Wen-Ji, and Yang Yu. 2020. *Temporal-Adaptive Hierarchical Reinforcement Learning*. https://doi.org/10.48550/arXiv.2002.02080.